Practice workbook · Identify blood components and match them to their features and jobs

Recognising red cells, white cells, platelets and plasma in smears or drawings, and tick grids or naming prompts pairing each with what it contains or does.

14 question parts: 1 worked-example retry and 13 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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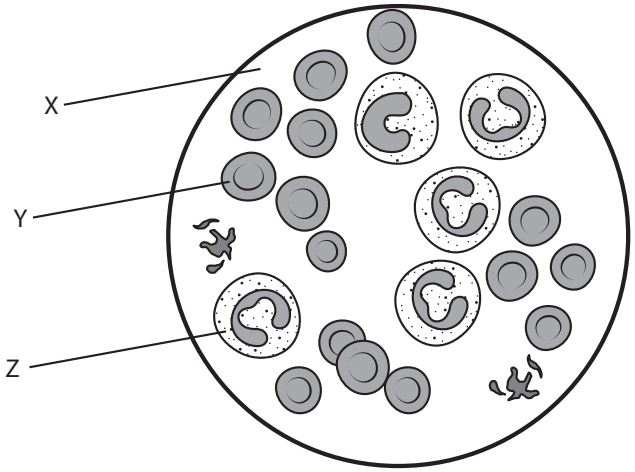
Type ID: HBI-BLOOD-COMPONENTS-IDENTIFY-AND-FUNCTION

# 1. Question 2 · part a.i · 3 marks

June 2024 Paper 1R

Worked-example retry · Attempt part a.i only.

**2** (a) The diagram shows components, X, Y and Z, in a sample of human blood.




(i) Name components X, Y and Z.

**(3)**

X .....

Y .....

Z .....

(ii) Describe the function of component Y.

**(2)**

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(iii) A sample of blood from a patient is analysed in a hospital.

A scientist finds that the number of component Z in the blood sample is higher than normal.

State a possible reason for the increase in the number of component Z.

**(1)**

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## 1. Question 2 · part a.i · 3 marks · Working and answer

Working and answer

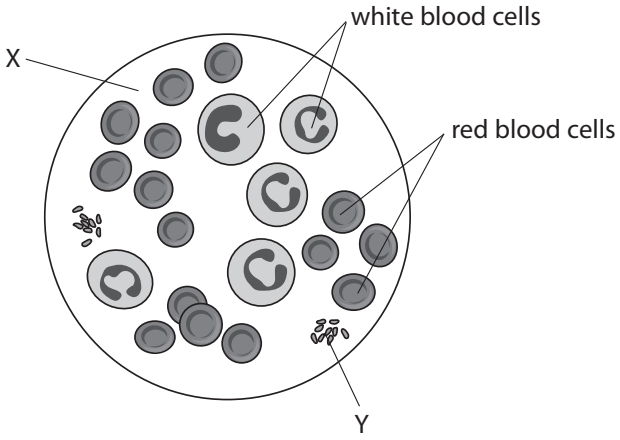
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## 2. Question 1 · part b · 2 marks

November 2020 Paper 1R

Further practice · Attempt part b only.

**1** The diagram shows a sample of human blood.




(a) State the function of red blood cells.

**(1)**

.....

.....

(b) Give the name of X and Y.

**(2)**

X .....

Y .....

(c) (i) Calculate the ratio of red blood cells to white blood cells.

Write the ratio in its simplest form.

**(2)**

ratio = .....

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## 2. Question 1 · part b · 2 marks · Working and answer

Working and answer

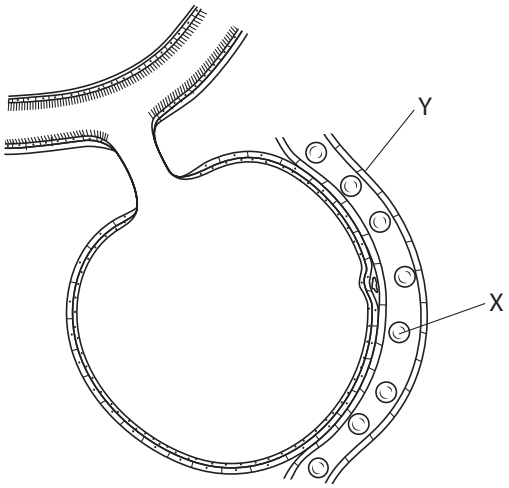
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# 3. Question 3 · part a.i · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.i only.

**3** The diagram shows an alveolus and its blood supply.




(a) (i) What is the name of blood component X? **(1)**

- ☒ **A** plasma
- ☒ **B** platelet
- ☒ **C** red blood cell
- ☒ **D** white blood cell

(ii) State three ways in which air in the alveolus differs from air in the atmosphere. **(3)**

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2 .....

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3 .....

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## 3. Question 3 · part a.i · 1 mark · Working and answer

Working and answer

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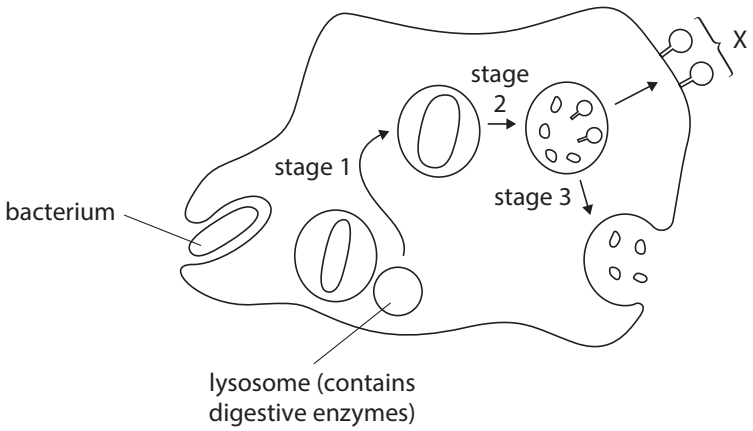
## 4. Question 6 · part b.ii · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.ii only.

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**6** The diagram shows a white blood cell engulfing a bacterium and the stages that follow.




(a) (i) State the name of the process used by white blood cells to engulf bacteria.

**(1)**

(ii) Describe what is happening at stages 1, 2 and 3 shown in the diagram.

**(3)**

stage 1

stage 2

stage 3

(iii) Suggest the function of the structures labelled X.

**(2)**

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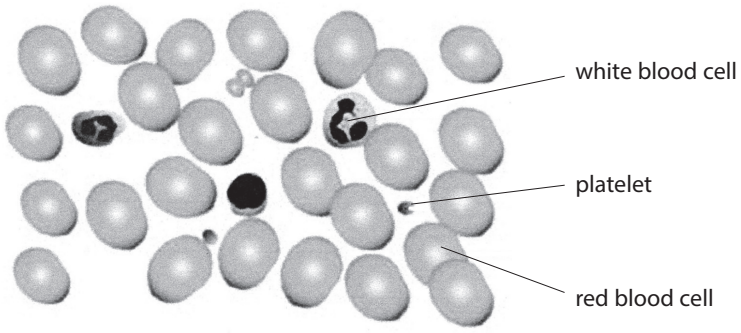
# 4. Question 6 · part b.ii · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.ii only.

(b) A student observes a prepared slide of blood under a light microscope.

The student draws this diagram of what could be seen.




(i) Describe how the slide of human blood could be prepared.

**(3)**

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# 4. Question 6 · part b.ii · 4 marks

June 2023 Paper 1

Further practice · Attempt part b.ii only.

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(ii) Discuss the accuracy of the student's diagram.

**(4)**

(iii) Explain why the student would be advised to use a prepared slide rather than making one from their own blood.

**(2)**

**(Total for Question 6 = 15 marks)**

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## 4. Question 6 · part b.ii · 4 marks · Working and answer

Working and answer

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## 5. Question 7 · part a · 2 marks

November 2020 Paper 2

Further practice · Attempt part a only.

**7** (a) Give two differences between whole blood and plasma.

**(2)**

1 .....

.....

2 .....

.....

(b) Ebola is an infectious disease. One symptom of Ebola is blood loss.

The World Health Organisation (WHO) has a procedure for treating Ebola.

This is the WHO's procedure.

- find a person who has recovered from Ebola
- observe this person to make sure that they are disease free for 28 days
- take blood from this person and test the blood for other diseases
- separate the plasma from the blood
- transfer this plasma to another person who has Ebola

Explain how the WHO procedure helps to treat a person who has Ebola.

**(6)**

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P 6 2 0 5 3 A 0 2 0 2 4

## 5. Question 7 · part a · 2 marks · Working and answer

Working and answer

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## 6. Question 2 · part a.iv · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.iv only.

**2** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

**Treatment of Ebola**

Ebola is caused by an RNA virus. There is no proven treatment available for Ebola. However, whole blood collected from someone who has recovered from infection has been used as a treatment. The results have been promising in a small group of Ebola patients.

Guidance for national health authorities and blood transfusion services describes the stages needed to collect whole blood (CWB) or plasma (CP) from Ebola-recovered patients. This can be used for transfusion to patients as a treatment for early Ebola.

The guidance states how to

- identify people who have recovered from Ebola as potential blood donors
- screen donors' blood for A, B, O and Rhesus groupings
- screen donors' blood for transfusion-transmissible infections
- collect blood and care for donors
- obtain agreement of Ebola patients for the treatment
- identify patient's blood grouping
- store and transport CWB and CP to the places where transfusions are to be given
- select Ebola patients for transfusion
- transfuse the blood into the patient

Donated CWB should never be frozen and should be stored between 2 °C and 6 °C.

CWB and CP donations should be transfused to the Ebola patients using standard transfusion procedures.

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## 6. Question 2 · part a.iv · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.iv only.

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(a) (i) Explain what is meant by an RNA virus (line 1).

**(2)**

(ii) The table lists five diseases. Put a tick (✓) next to the diseases that should be tested for in the donors' blood.

**(2)**

| Disease | Blood tested (✓) |
|---|---|
| anaemia | |
| cystic fibrosis | |
| gonorrhoea | |
| HIV | |
| scurvy | |

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(iii) Explain why it is necessary to identify the blood group of the blood collected from donors who have recovered from Ebola (line 9).

**(3)**

## 6. Question 2 · part a.iv · 2 marks

November 2020 Paper 2R

Further practice · Attempt part a.iv only.

(iv) Suggest why donated CWB should never be frozen (line 17).

**(2)**

(v) Explain how the blood of people who have recovered from Ebola is likely to be effective in treating the disease.

**(2)**

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(b) The nitrogenous base composition of another virus was found to be adenine 13%, guanine 26%, uracil 20% and cytosine 41%.

Explain how these figures prove that this is a virus with a single strand of RNA.

**(3)**

**(Total for Question 2 = 14 marks)**

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## 6. Question 2 · part a.iv · 2 marks · Working and answer

Working and answer

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## 7. Question 1 · part whole · 6 marks

June 2019 Paper 1R

Further practice · Attempt part whole only.

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**1** The table lists blood components and their features.

Place ticks (✓) in boxes to show the features of each blood component.

**(6)**

| | Feature | | | | | |
|---|---|---|---|---|---|---|
| **Blood component** | **has a nucleus** | **contains haemoglobin** | **carries hormones** | **engulfs pathogens** | **produces antibodies** | **helps blood clotting** |
| red blood cell | | | | | | |
| phagocyte | | | | | | |
| lymphocyte | | | | | | |
| platelets | | | | | | |
| plasma | | | | | | |

**(Total for Question 1 = 6 marks)**

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## 7. Question 1 · part whole · 6 marks · Working and answer

Working and answer

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## 8. Question 1 · part whole · 4 marks

June 2021 Paper 1

Further practice · Attempt part whole only.

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**1** The table lists four features of blood. It also shows four main components of blood.

Put a tick (✓) in a box if the component shows the feature.

**(4)**

| Feature | Components | | | |
|---|---|---|---|---|
| | **Red blood cell** | **Plasma** | **Platelet** | **Phagocyte** |
| transports oxygen | | | | |
| has a nucleus | | | | |
| consists of 90% water | | | | |
| involved in blood clotting | | | | |

**(Total for Question 1 = 4 marks)**

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Working and answer

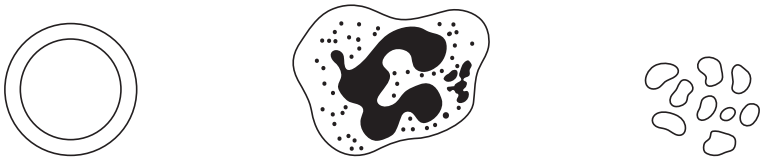
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# 9. Question 3 · part a · 4 marks

November 2021 Paper 1

Further practice · Attempt part a only.

**3** The diagram shows some of the components of blood.



A B C

(a) Complete the table by giving the name of the component and its function.

Component A has been completed as an example.

**(4)**

| Component | Name | Function |
|---|---|---|
| **A** | red blood cell | transports oxygen |
| **B** | | |
| **C** | | |

(b) (i) Describe the process of blood clotting.

**(5)**

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## 9. Question 3 · part a · 4 marks · Working and answer

Working and answer

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## 10. Question 1 · part a · 5 marks

November 2023 Paper 1

Further practice · Attempt part a only.

**1** (a) The table gives components of the blood and their features.

Complete the table by putting a tick (✓) in the box if the blood component shows the feature.

**(5)**

| Feature | Blood component | | | | |
|---|---|---|---|---|---|
| | red blood cell | phagocyte | lymphocyte | platelet | plasma |
| transports oxygen | | | | | |
| contains haemoglobin | | | | | |
| is a fluid | | | | | |
| produces antibodies | | | | | |
| engulfs pathogens | | | | | |

(b) Explain why it is important that blood clots when the skin is cut open.

**(4)**

**(Total for Question 1 = 9 marks)**

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## 10. Question 1 · part a · 5 marks · Working and answer

Working and answer

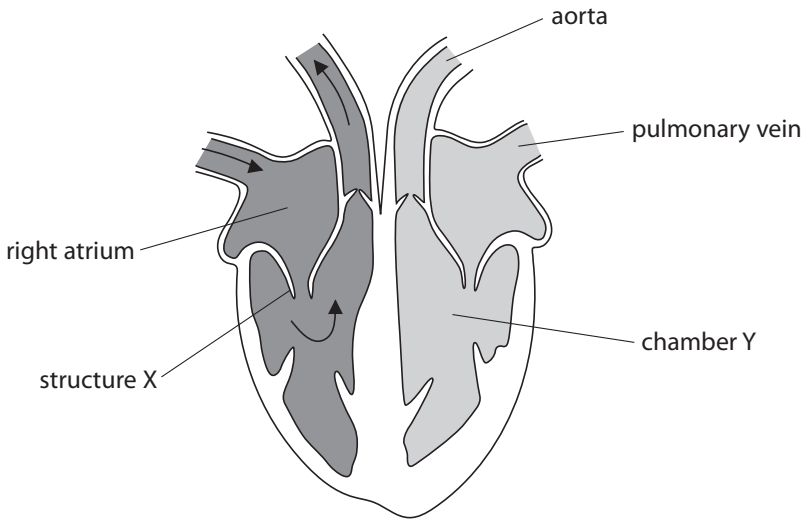
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# 11. Question 2 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

**2** (a) The diagram shows a section through the heart.




(i) Name chamber Y. **(1)**

(ii) Explain the function of structure X. **(2)**

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# 11. Question 2 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

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(iii) The table lists some properties of blood.

Add ticks (✓) to those boxes that show the properties of blood in the aorta.

**(2)**

| Properties of blood | Blood in the aorta |
|---|---|
| oxygenated | |
| deoxygenated | |
| high pressure | |
| medium pressure | |
| low pressure | |

(iv) State where blood in the left side of the heart has come from.

**(1)**

(b) In one year in the United Kingdom, there were 173 heart transplants.

Heart transplant operations require hearts to be donated.

Some hearts cannot be used because the donor is too old.

Give two other reasons why a heart might not be suitable for a transplant.

**(2)**

1 .....

.....

2 .....

.....

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# 11. Question 2 · part c.i · 1 mark

January 2023 Paper 2

Further practice · Attempt part c.i only.

(c) (i) Which blood component causes blood clotting?

(1)

- ☒ **A** lymphocytes
- ☒ **B** phagocytes
- ☒ **C** platelets
- ☒ **D** red blood cells

(ii) The diagram summarises part of the process that causes a blood clot.




Anticoagulant drugs prevent a blood clot forming.

These drugs stop the enzyme thrombin being produced.

Describe how a lack of thrombin prevents a blood clot forming.

(2)

**(Total for Question 2 = 11 marks)**

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## 11. Question 2 · part c.i · 1 mark · Working and answer

Working and answer

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# 12. Question 5 · part c · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c only.

**5** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The heart is a double-pump that is part of the human circulatory system. It helps ensure that the blood and its components reach all cells in the body.

Some components of the blood are made in the bone marrow. An average adult with a mass of 70 kg has 5.5 $dm^3$ of blood. About 45% of the blood is made up of different cells. There are several different types of cell in the blood but red blood cells are the most common. In males, 1.0 $dm^3$ of blood contains $5.9 \times 10^{12}$ red blood cells. In females the same volume of blood contains $5.5 \times 10^{12}$ red blood cells. Haemoglobin makes up a large part of red blood cells. This protein is vital to the function of red blood cells.

Red blood cells are transported around the body in a liquid known as plasma. Only 10% of blood plasma is made up of components that are used or produced by body cells. The rest of the plasma is made up of water. Some components of blood plasma are used by body cells in chemical reactions, some act as messengers or have a role in blood clotting and others are excreted from the body as waste.

(a) Name the main tissue that makes up the walls of the heart. **(1)**

(b) Explain why the heart is known as a double-pump. (Lines 1 to 2) **(3)**

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## 12. Question 5 · part c · 1 mark

June 2024 Paper 2R

Further practice · Attempt part c only.

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(c) Red blood cells are the most common type of cell found in the blood.

Which of these are also types of cell found in the blood?

**(1)**

- ☒ **A** phagocytes and platelets
- ☒ **B** antibodies and hormones
- ☒ **C** lymphocytes and phagocytes
- ☒ **D** platelets and lymphocytes

(d) (i) Name a component of the blood plasma that acts as a messenger. (Lines 12 to 14)

**(1)**

.....

(ii) Name a component of the blood that is involved in blood clotting.

**(1)**

.....

(e) Calculate the number of red blood cells found in the blood of an adult male with a mass of 95 kg. (Lines 3 to 7)

Give your answer in standard form.

**(3)**

number of red blood cells = .....

**(Total for Question 5 = 10 marks)**

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## 12. Question 5 · part c · 1 mark · Working and answer

Working and answer

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# 13. Question 5 · part d.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part d.ii only.

**5** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

The heart is a double-pump that is part of the human circulatory system. It helps ensure that the blood and its components reach all cells in the body.

Some components of the blood are made in the bone marrow. An average adult with a mass of 70 kg has 5.5 $dm^3$ of blood. About 45% of the blood is made up of different cells. There are several different types of cell in the blood but red blood cells are the most common. In males, 1.0 $dm^3$ of blood contains $5.9 \times 10^{12}$ red blood cells. In females the same volume of blood contains $5.5 \times 10^{12}$ red blood cells. Haemoglobin makes up a large part of red blood cells. This protein is vital to the function of red blood cells.

Red blood cells are transported around the body in a liquid known as plasma. Only 10% of blood plasma is made up of components that are used or produced by body cells. The rest of the plasma is made up of water. Some components of blood plasma are used by body cells in chemical reactions, some act as messengers or have a role in blood clotting and others are excreted from the body as waste.

(a) Name the main tissue that makes up the walls of the heart. **(1)**

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# 13. Question 5 · part d.ii · 1 mark

June 2024 Paper 2R

Further practice · Attempt part d.ii only.

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(c) Red blood cells are the most common type of cell found in the blood.

Which of these are also types of cell found in the blood?

**(1)**

- ☒ **A** phagocytes and platelets
- ☒ **B** antibodies and hormones
- ☒ **C** lymphocytes and phagocytes
- ☒ **D** platelets and lymphocytes

(d) (i) Name a component of the blood plasma that acts as a messenger. (Lines 12 to 14)

**(1)**

.....

(ii) Name a component of the blood that is involved in blood clotting.

**(1)**

.....

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Give your answer in standard form.

**(3)**

number of red blood cells = .....

**(Total for Question 5 = 10 marks)**

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## 13. Question 5 · part d.ii · 1 mark · Working and answer

Working and answer

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# 14. Question 5 · part d.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part d.i only.

**5** Read the passage below. Use the information in the passage and your own knowledge to answer the questions that follow.

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Red blood cells are transported around the body in a liquid known as plasma. Only 10% of blood plasma is made up of components that are used or produced by body cells. The rest of the plasma is made up of water. Some components of blood plasma are used by body cells in chemical reactions, some act as messengers or have a role in blood clotting and others are excreted from the body as waste.

(a) Name the main tissue that makes up the walls of the heart. **(1)**

(b) Explain why the heart is known as a double-pump. (Lines 1 to 2) **(3)**

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# 14. Question 5 · part d.i · 1 mark

June 2024 Paper 2R

Further practice · Attempt part d.i only.

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(c) Red blood cells are the most common type of cell found in the blood.

Which of these are also types of cell found in the blood?

**(1)**

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- ☒ **B** antibodies and hormones
- ☒ **C** lymphocytes and phagocytes
- ☒ **D** platelets and lymphocytes

(d) (i) Name a component of the blood plasma that acts as a messenger. (Lines 12 to 14)

**(1)**

.....

(ii) Name a component of the blood that is involved in blood clotting.

**(1)**

.....

(e) Calculate the number of red blood cells found in the blood of an adult male with a mass of 95 kg. (Lines 3 to 7)

Give your answer in standard form.

**(3)**

number of red blood cells = .....

**(Total for Question 5 = 10 marks)**

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## 14. Question 5 · part d.i · 1 mark · Working and answer

Working and answer

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